



Hoppers Challenge: Sustainable transport

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A sustainable world is one in which our environmental systems and societies flourish, natural resources are able to replenish themselves, and people are able to live healthy and secure lives. The growing complexity and scale of scientific computing workflows and AI training pipelines place huge demands on energy and other natural resources. Sustainability can be overlooked in favour of technological advancements, therefore it is of utmost importance for us to make the effort to consider how we can fit sustainability into our own projects.

The Challenge

Consider the fact that Jane street workers usually travel by car. Make a program that encourages workers to take an alternative and more sustainable route to the office.

Examples

- Think about gamifying actions
- How could you convince employees to walk or take the bus?
- How this can benefit the employee?

Why is sustainable transport important?

Sustainable transport aims to minimize the emissions of vehicles as the transport sector is responsible for 13.7% of global greenhouse gas emissions. This may be through setting up more public transport, technological advancements in vehicles (i.e. electric cars) or encouraging people to not use a vehicle at all (i.e. walking, cycling). Sustainable transport also aids in financial security as it means people don't have to be reliant on volatile and ever increasing energy/fuel prices.

How will we judge?

Marks will be given for the following.

- Teamwork (how much did everyone contribute)
- Originality
- Theme (How well does your project fit the theme of Sustainability?)
- Presentation Skills
- Project Quality



Extra Resources

<https://www.gov.uk/government/collections/government-conversion-factors-for-company-reporting>

<https://data.bus-data.dft.gov.uk/>

<https://www.nomisweb.co.uk/>

<https://github.com/Project-OSRM/osrm-backend>

<https://open-meteo.com/>

Good luck!